

No Straight Surfaces

While traditional curved surfaces are good reflectors as well, radar reflects very well off straight surfaces, almost directly back to the source. To minimise that, either many differently arranged straight surfaces are used to break the reflection into several directions like in the B-2, or, specially curved surfaces like F-22s.

Super-Cruise Engines

Surely you haven't given a thought to how a plane takes off, flies and lands without tripping over things, even while you were jet setting off to some destination! Well, the entire process is controlled by what is known as the air traffic control (ATC) system.

Angled Jet Exhaust

Minimising the area of exhaust heat from jet engine, the exhausts are designed to direct the low. Stealth bombers, like the B-2, also lower their infrared signature by angling the exhaust over the body of the plane.

Composite, Super-Strong Materials

Metal is a good reflector of radio waves, whereas carbon, and compounds such as plastic are bad reflectors. Composite material are therefore used to create superstrong materials using metals, carbon, and more in the F-22 Raptor's build design.

Canted Tail Fins

Tail fins provide an excellent broadside radar cross-section, if mounted conventionally. Canted tail fins will redirect the reflection. The entire process is controlled by the air traffic control (ATC) system.

Thrust Vectoring

While this is not truly stealth technology, and not a must on a stealth plane, its applications in manoeuvrability and exhaust control are numerous.

